

3. Let $A = \left\{ x \in \mathbb{R} \mid \frac{x^2 + 12}{x} < 7 \right\}$, $B = \left\{ x \in \mathbb{R} \mid \frac{x}{x+2} \geq \frac{1}{x} \right\}$. Find $A \cap B'$. (4 points)

4. Let $D_{r^{-1}} = [0, \infty)$ and $D_r = (-\infty, -2) \cup [3, \infty)$ when minimum point is $(3, 0)$

Find relation r and r^{-1} . (Answer in a Set-builder notation form.)

(3 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม

7. Let $f(x-2) = \frac{x^2-9}{x-5}$, $(f \cdot g)(x) = x^3 - 2x^2 - 15x$ and $h^{-1}\left(\frac{1}{x+1}\right) = \sqrt{x-1}$.

Find $\left(\frac{g}{h}\right)(x)$ and $D_{\frac{g}{h}}$. (4 points)

วิทยาลัยเทคโนโลยีอุตสาหกรรม